Optimization of Biomass Valorisation Networks for a Climate-Neutral and Sustainable Economy: A Multi-Objective Modelling Approach

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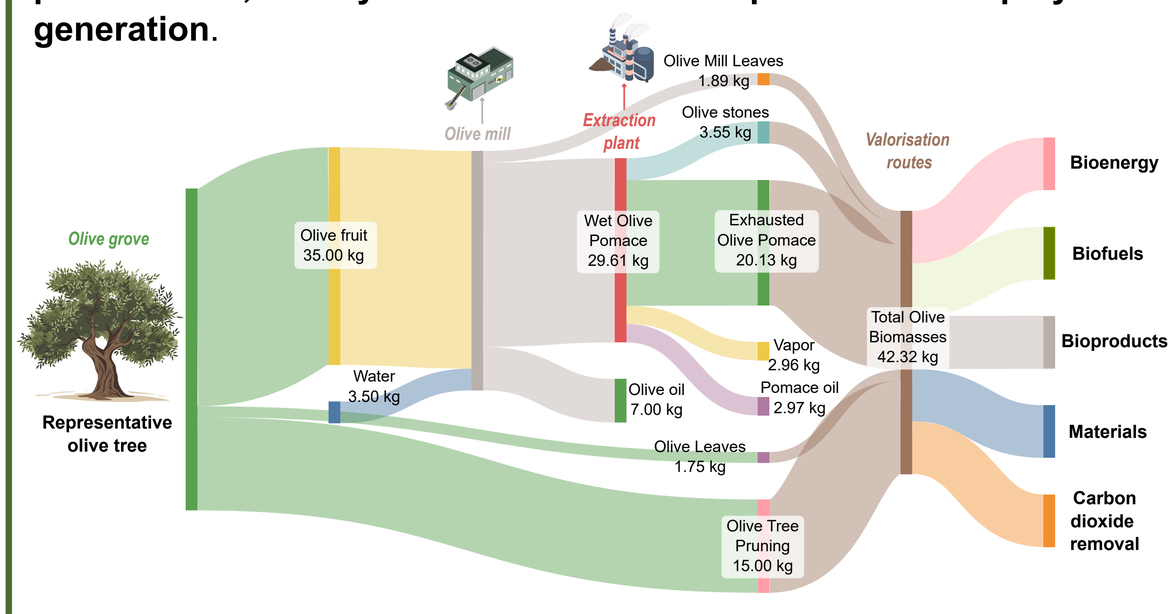
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INTRODUCTION

The transition towards a **climate-neutral bioeconomy** requires the **strategic allocation** of **biomass** resources across competing valorisation pathways. **Olive cultivation and oil production** generate substantial and heterogeneous biomass streams **with applications** in bioenergy, biofuels, bioproducts, materials and carbon dioxide removal. However, **identifying their most sustainable use** requires moving beyond individual conversion technologies towards integrated decision-support frameworks that simultaneously consider **economic performance, life-cycle environmental impacts and employment generation**.



OLIVE BIOMASS-TO-WHAT?

How should scarce biogenic carbon be allocated to maximise sustainability benefits?

METHODOLOGY

Network-based optimization model

A **network-based linear programming model** was developed to identify **optimal valorisation portfolios** for olive-derived biomass in **Andalusia (Spain)**. The superstructure integrates biomass feedstocks, alternative conversion technologies, intermediate streams, final products, conventional supply routes and carbon dioxide removal (CDR) pathways, including Bio-CCS and biochar.

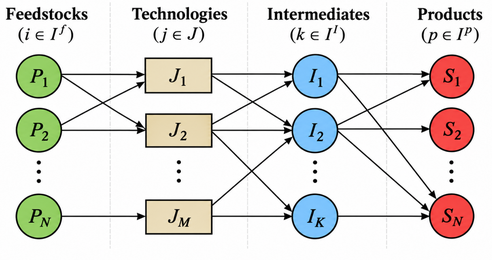
5 *Olive-derived biomass streams*

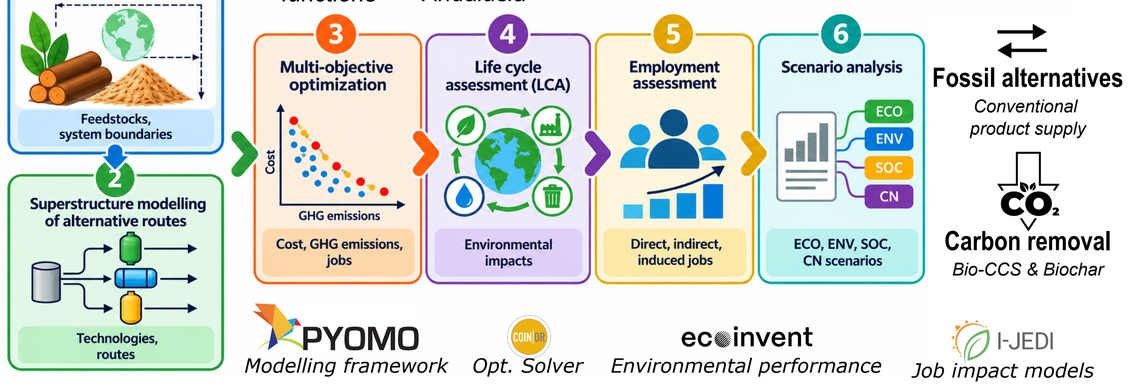
78 *Conversion technologies*

8 *Final products*

3 *Objective functions*

1 *Regional case Andalusia*





PYOMO *Modelling framework* · COIN-OR *Opt. Solver* · ecoinvent *Environmental performance* · I-JEDI *Job impact models*

Sustainability scenarios analyzed

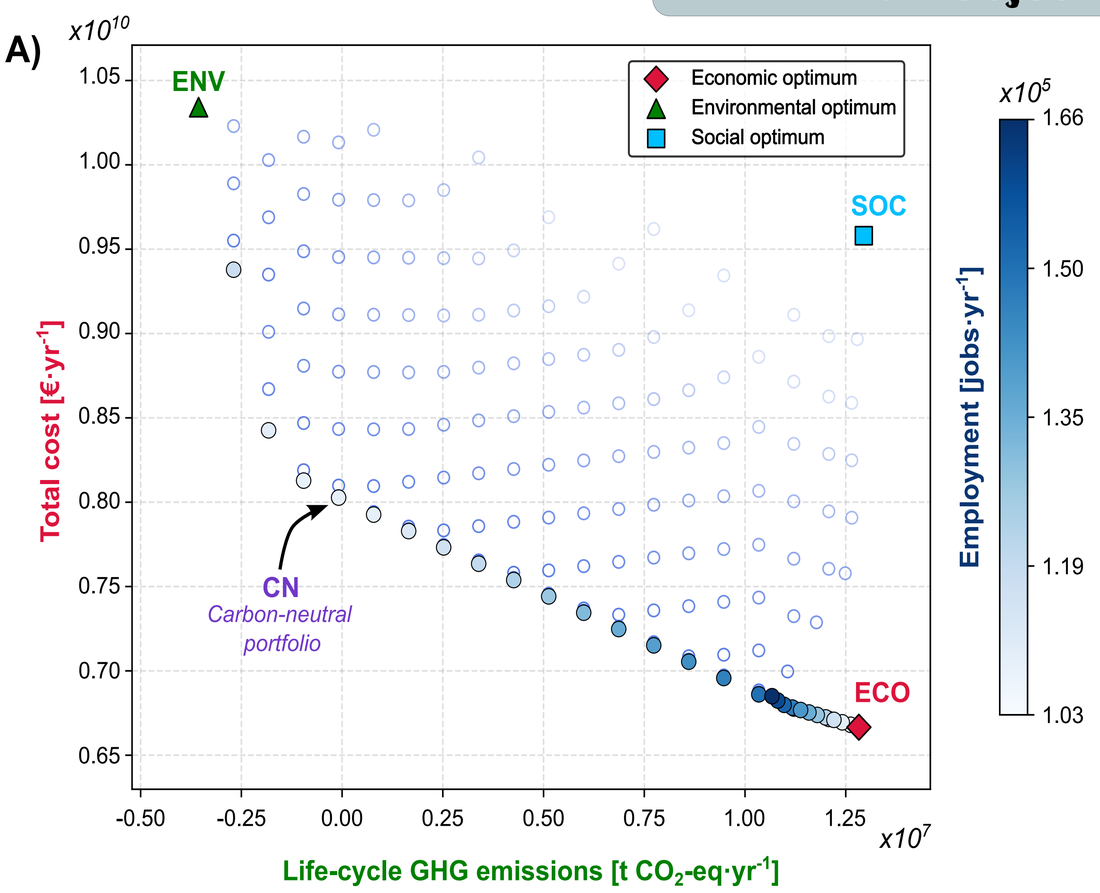
Economic optimum — *Minimize total cost*

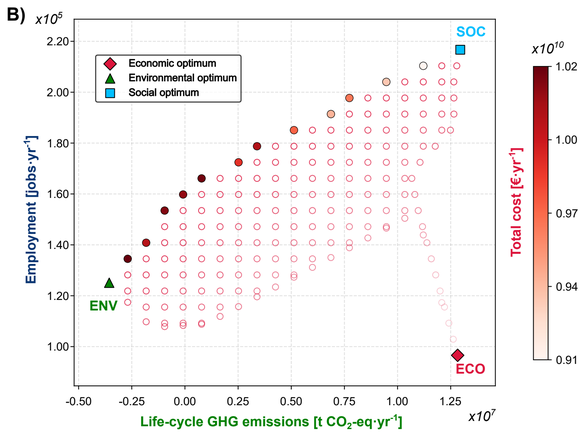
Environmental optimum — *Minimize GHG emissions*

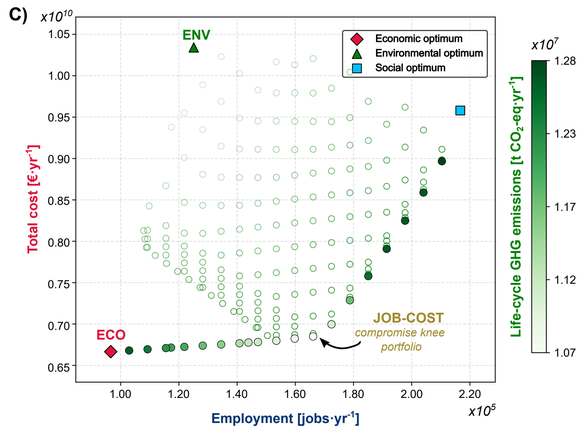
Social optimum — *Maximize employment*

RESULTS & DISCUSSION

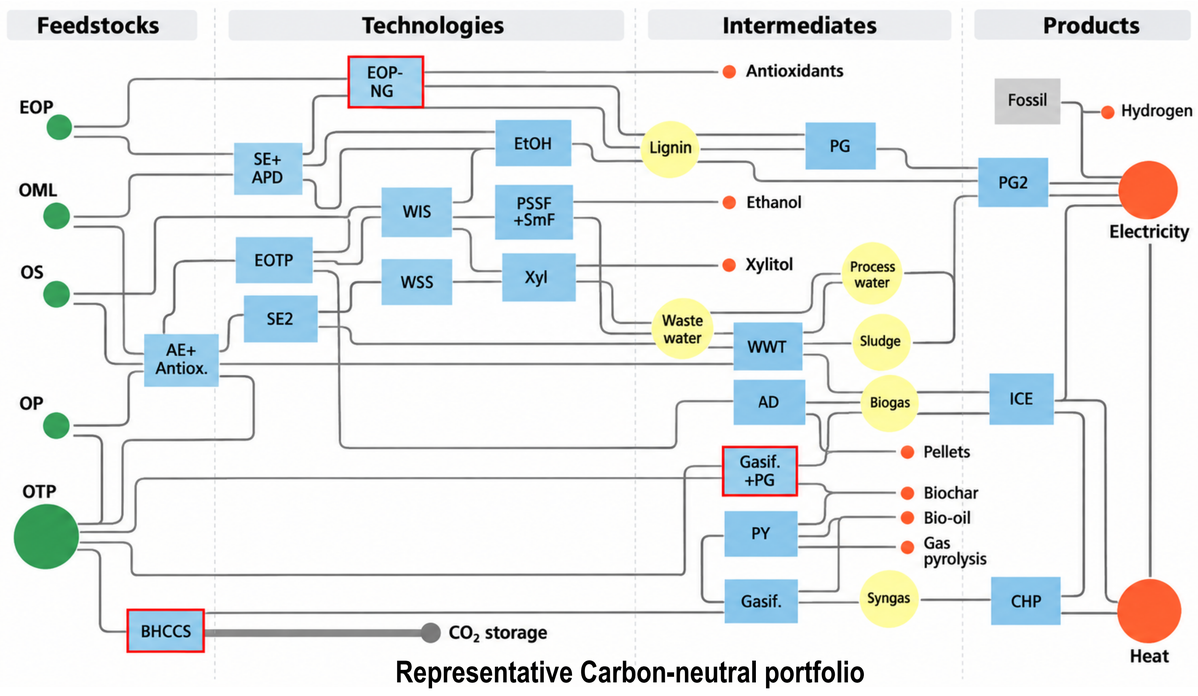
Multiobjective Pareto Frontiers







Representative optimal network configuration



Representative Carbon-neutral portfolio

Key findings

-3.61 Mt
CO_2-eq yr^{-1}
Environmental optimum

+21.5%
System cost
Near-carbon neutrality vs. ECO

2.35x
Employment
SOC relative to ECO

HEAT
Main switching product
Towards BIO-CCS supply

CONCLUSIONS

1. Optimal **biomass allocation** strongly **depends on the sustainability objective** considered.
2. **Carbon neutrality** is primarily driven by **selective deployment of Bio-CCS heat**.
3. Employment-oriented biomass deployment does **not automatically deliver climate mitigation**.
4. Targeted routes combined with **durable carbon storage** provide the highest climate value.

ACKNOWLEDGEMENTS

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